Video Transcript

Video 10: Using Naïve Bayes Theorem to Calculate Probabilities for Multiple Features (04:27)

Now, so far we've just assumed that we're only looking at one word in the email. So, that's not very realistic. The email will have many words in it. And so, the question is now, what do we do when we have lots of words? So, let's consider the case where we have two words. We have money and lunch. And so, we want to understand what's the probability of this email being normal or spam. So, one thing that we could do is we could say, well, given that this is a normal e-mail, what's the probability that money, that feature money is in the email and what's the probability that lunch is in that email? Now, if we assume that these probabilities are totally independent, and we'll see that that's typically what we will do, we'll assume that each feature occurs independently of the other.

So, we assume that, that feature is being drawn from an IID distribution, an independent identically distributed distribution. So, if they are independent, then we can just multiply the probabilities. Given that this is a normal email, the probability that feature one occurs, we can multiply by the probability that feature two occurs. And so for money and lunch, we would just do this calculation of these probabilities being multiplied. And what we'd see is that the probability, the relative probability of it being normal is 1/10. So, the conclusion would be that this is spam.

Now this assumption, that things are independent of each other is given the term Naive Bayes. And, what it means is that if we have a number of features, we can just multiply their probabilities by each other. So, here is the new equation where if we've got N features, then we have a product of N conditional probability distributions. So, probability of f1 given N times the probability of f2 given N, etc. So, you can imagine if we have 1,000 words, we've got a lot of multiplications here and these probabilities are all less than one and we can run into some difficulties computationally with underflow.

Now, we can get around that by instead of dealing with the probabilities, we can take the log of

© MIT xPRO® 2024. All rights reserved.

the probability and that's often done. And, so taking just either the numerator or the denominator here, we can express it in this way that taking the logs, then this becomes a sum. We use the property that log of A times B is equal to the log of A plus the log of B. So, now we have a summation which computationally is much more friendly.

Now, there's lots of other issues that will arise in this kind of computation, so if you want to understand more, I suggest you start reading at this link here, which takes you through what's called the maximum a posteriori for machine learning. Basically, it's dealing with these probability distributions and how we sample them. For data engineering, we may need to know some of these tricks, but not all of them. So, now why don't you calculate the posterior relative probability for an email with the words password and the money in it? Okay, bye for now.

© MIT xPRO® 2024. All rights reserved.